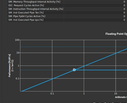

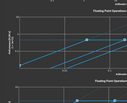

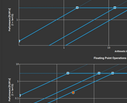

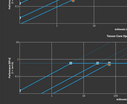

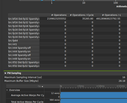

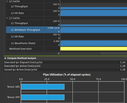

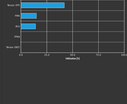

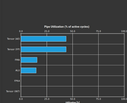

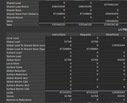

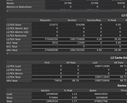

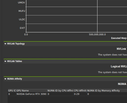

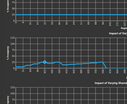

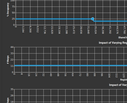

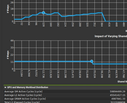

Result	Size	Time	Cycles	GPU	SM Frequency	Process	Attributes
Current 686 - flash_attention	(1024, 1, 1)x(128, 1, 1)	44.76 ms	62,398,567	0 - NVIDIA GeForce RTX 3090	1.39 Ghz	[125804] python3.12	

Summary
Details
Source
Context
Comments
Raw
Session
Compare
Tools
View
Export
All

GPU Speed Of Light Throughput

Compute (SM) Throughput [%]	42.00	Duration [ms]	44.76
Memory Throughput [%]	36.49	Elapsed Cycles [cycle]	62398567
L1/TEX Cache Throughput [%]	26.57	SM Active Cycles [cycle]	59894499.29
L2 Cache Throughput [%]	36.49	SM Frequency [Ghz]	1.39
DRAM Throughput [%]	1.17	DRAM Frequency [Ghz]	9.49

GPU Throughput

Compute (SM) [%]
Memory [%]
0.0
10.0
20.0
30.0
40.0
50.0
60.0
70.0
80.0
90.0
100.0
Speed Of Light (SOL) [%]

Compute Throughput Breakdown

SM: Pipe Tensor Cycles Active [%]	42.00
SM: Issue Active [%]	28.43
SM: Inst Executed [%]	28.43
SM: Inst Executed Pipe Lsu [%]	21.14
SM: Pipe Fmaheavy Cycles Active [%]	17.01
SM: Pipe Fma Cycles Active [%]	14.97
SM: Pipe Alu Cycles Active [%]	14.13
SM: Mio2rf Writeback Active [%]	11.87
SM: Mio Inst Issued [%]	7.52
SM: Mio Pq Write Cycles Active [%]	2.64
SM: Mio Pq Read Cycles Active [%]	2.64
SM: Inst Executed Pipe Adu [%]	1.42
SM: Inst Executed Pipe Cbu Pred On Any [%]	0.66
SM: Inst Executed Pipe Uniform [%]	0.33
SM: Inst Executed Pipe Xu [%]	0.00
SM: Memory Throughput Internal Activity [%]	0
IDC: Request Cycles Active [%]	0
SM: Instruction Throughput Internal Activity [%]	0
SM: Inst Executed Pipe Tex [%]	0
SM: Pipe Fp64 Cycles Active [%]	0
SM: Inst Executed Pipe Ipa [%]	0

Memory Throughput Breakdown

L2: T Sectors [%]	36.49
L2: Lts2xbar Cycles Active [%]	36.23
L1: Data Pipe Lsu Wavefronts [%]	25.52
L1: Lsu Writeback Active [%]	23.69
L1: Lsuin Requests [%]	21.14
L1: M Xbar2l1tex Read Sectors [%]	21.01
L1: Data Bank Reads [%]	10.50
L2: D Sectors [%]	9.19
L2: T Tag Requests [%]	9.12
GPU: Compute Memory Access Throughput Internal Activity [%]	9.07
L2: Xbar2lts Cycles Active [%]	4.55
L1: M L1tex2xbar Req Cycles Active [%]	2.64
L1: Data Bank Writes [%]	2.63
DRAM: Cycles Active [%]	1.17
DRAM: Dram Sectors [%]	1.01
L2: D Sectors Fill Device [%]	0.49
L1: Texin Sm2tex Req Cycles Active [%]	0.16
L2: D Sectors Fill Sysmem [%]	0.00
L1: Tex Writeback Active [%]	0
L2: D Atomic Input Cycles Active [%]	0
L1: F Wavefronts [%]	0
L1: Data Pipe Tex Wavefronts [%]	0
GPU: Compute Memory Request Throughput Internal Activity [%]	0

Floating Point Operations Roofline

100
10
1
0.1
Performance [FLOP/s] (1 = 1e+12)
0.1
1
10
100
1,000
10,000
100,000
Arithmetic Intensity [FLOP/byte]

Floating Point Operations Roofline (Double Precision)

10
1
0.1
Performance [FLOP/s] (1 = 1e+12)
0.01
0.1
1
10
100
Arithmetic Intensity [FLOP/byte]

Floating Point Operations Roofline (Half Precision)

10
1
Performance [FLOP/s] (1 = 1e+12)
1
10
100
1,000
10,000
Arithmetic Intensity [FLOP/byte]

Floating Point Operations Roofline (Single Precision)

10
1
0.1
Performance [FLOP/s] (1 = 1e+12)
1
10
100
1,000
10,000
100,000
Arithmetic Intensity [FLOP/byte]

Tensor Core Operations Roofline

10
1
0.1
Performance [OP/s] (1 = 1e+14)
1
10
100
1,000
10,000
100,000
1e+06
Arithmetic Intensity [OP/byte]

	# Operations	# Operations / Cycle	# Operations / s	Peak %	Peak Operations / Cycle	Peak Operations / s
Src:bf16 Dst:fp32 Sparsity:	2199023255552	35265.80	49128960023792.55	84.00	41984	58488119471788.16
Src:bf16 Dst:fp32 Sparsity:	0	0	0	0	83968	116976238943576.33
Src:fp16 Dst:fp16 Sparsity:	0	0	0	0	83968	116976238943576.33
Src:fp16 Dst:fp16 Sparsity:	0	0	0	0	167936	233952477887152.66
Src:fp16 Dst:fp32 Sparsity:	0	0	0	0	41984	58488119471788.16
Src:fp16 Dst:fp32 Sparsity:	0	0	0	0	83968	116976238943576.33
Src:fp64	0	0	0	0	289.54	403366341184.75
Src:int1	0	0	0	0	1343488	1871619823097221...
Src:int4 Sparsity:off	0	0	0	0	335872	467904955774305.31
Src:int4 Sparsity:on	0	0	0	0	671744	935809911548610.62
Src:int8 Sparsity:off	0	0	0	0	167936	233952477887152.66
Src:int8 Sparsity:on	0	0	0	0	335872	467904955774305.31
Src:tf32 Dst:fp32 Sparsity:	0	0	0	0	20992	29244059735894.08
Src:tf32 Dst:fp32 Sparsity:	0	0	0	0	41984	58488119471788.16

PM Sampling

Maximum Sampling Interval [us]	16	# Pass Groups	2
Maximum Buffer Size [Mbyte]	28.18		-

+5ms
+10ms
+15ms
+20ms
+25ms
+30ms
+35ms
+40ms
+45ms
Overview
Average Active Warps Per Cy
12 warp
0
Total Active Warps Per Cycle
984 warp
0
Blocks Launched
40 block
0
SM Active Cycles
22.3k cycle
0
Executed Ipc Active
...6 inst/cycle
0
SM
SM ALU Pipe Throughput
100 %
0
SM FMA Light Pipe Throughp
SM FMA Heavy Pipe Through
SM Tensor Pipe Throughput
SM XU Pipe Throughput
DRAM
DRAM Throughput
DRAM Read Bandwidth
DRAM Write Bandwidth
L2 Cache
L2 Throughput
L2 Hit Rate
L1 Cache
L1 Throughput
L1 Writeback Throughput
5.96k cycle
L1 Hit Rate
L1 Wavefronts (Data)
6.46k
Workload Execution
flash_attention

Compute Workload Analysis

Executed Ipc Elapsed [inst/cycle]	1.14	SM Busy [%]	42.00
Executed Ipc Active [inst/cycle]	1.18	Issue Slots Busy [%]	28.43
Issued Ipc Active [inst/cycle]	1.18		

Pipe Utilization (% of elapsed cycles)

Tensor (All)
Tensor (FP)
FMA
ALU
FP64
Tensor (INT)
0.0
25.0
50.0
75.0
100.0
Utilization [%]

Pipe Utilization (% of peak instructions executed over elapsed cycles)

LSU
Tensor (FP)
FMA
ALU
ADU
Uniform
CBU
XU
FMA (FP16)
FP64 (DMMA)
FP64 (FP64)
TEX
Tensor (INT)
0.0
25.0
50.0
75.0
100.0
Utilization [%]

Pipe Utilization (% of active cycles)

Tensor (All)
Tensor (FP)
FMA
ALU
FP64
Tensor (INT)
0.0
25.0
50.0
75.0
100.0
Utilization [%]

Pipe Utilization (% of peak instructions executed over active cycles)

LSU
Tensor (FP)
FMA
ALU
ADU
Uniform
CBU
XU
FMA (FP16)
FP64 (DMMA)
FP64 (FP64)
TEX
Tensor (INT)
0.0
25.0
50.0
75.0
100.0
Utilization [%]

Memory Workload Analysis

Memory Throughput [Gbyte/s]	10.70	Mem Busy [%]	36.49
L1/TEX Hit Rate [%]	0	Max Bandwidth [%]	36.23
L2 Hit Rate [%]	98.67	Mem Pipes Busy [%]	21.14
L2 Compression Input Sectors [sector]	0	Local Memory Spilling Requests	0
L2 Compression Ratio	0	Local Memory Spilling Request Overhead [%]	0
L2 Compression Success Rate [%]	0	L2 Persisting Size [Mbyte]	1.18

Memory Chart

Values: Throughput
Inactivity: Greyed Out

Kernel
65.54 K Inst
Global
32.77 K Req
32.77 K Req
0.00 Inst
Local
0.00 Req
0.00 Req
0.00 Inst
Texture
0.00 Req
0.00 Inst
Surface
0.00 Req
0.00 Req
67.11 M Inst
Load Global Store Shared
0.00 B/s
268.44 M Inst
Shared
268.44 M Req
0.00 Req
Shared Memory
L1/TEX Cache Hit Rate: 0.00 %
768.01 GB/s
374.82 MB/s
767.64 GB/s
L2 Cache Hit Rate: 98.67 %
0.00 B/s
0.00 B/s
System Memory
10.29 GB/s
411.05 MB/s
Device Memory
0.00 B/s
0.00 B/s
Peer Memory
0.00 B/s
0.00 B/s
L2 Compression Ratio: 0.00
385.28 KB
699.49 M Req
ICC Hit Rate: 100.00 %
16.69 K Req
2.84 M Req
IMC Hit Rate: 99.91 %
984.00 Req
GCC Hit Rate: 96.29 %
0.00 Req
IDC Hit Rate: 0.00 %
% Peak
100%
80%
60%
40%
20%
0%

Shared Memory

	Instructions	Requests	Wavefronts	% Peak	Bank Conflicts
Shared Load	0	0	1074492242	21.01	26136
Shared Load Matrix	268435456	268435456			
Shared Store	0	0	0	0	0
Shared Store From Global Lo	67108864	67108864	268435456	1.31	0
Shared Atomic	0	0	0	0	0
Other	-	-	196408450	3.84	0
Total	335544320	335544320	1539336148	26.17	26136

L1/TEX Cache

	Instructions	Requests	Wavefronts	% Peak	Sectors	Sectors/Req	Hit Rate	Bytes	Sector
Local Load	0	0	0	0	0	0	0	0	
Global Load	32768	32768	134283264	2.63	524288	16	0	16777216	
Global Load To Shared Store (acce	0	0			0	0		0	
Global Load To Shared Store (bypa	67108864	67108864			1073741824	16	-	34359738368	
Surface Load	0	0	0	0	0	0	0	0	
Texture Load	0	0	0	0	0	0	0	0	
Global Store	32768	32768	65550	0.00	524288	16	0	16777216	
Local Store	0	0	0	0	0	0	0	0	
Surface Store	0	0	0	0	0	0	0	0	
Global Reduction	0	0	0	0	0	0	0	0	
Surface Reduction	0	0	0	0	0	0	0	0	
Global Atomic ALU	0	0	0	0	0	0	0	0	
Global Atomic CAS	0								
Surface Atomic ALU	0	0	0	0	0	0	0	0	
Surface Atomic CAS	0								
Loads	67141632	67141632	134283264	2.63	1074266112	16	0	34376515584	
Stores	32768	32768	65550	0.00	524288	16	0	16777216	
Atomics & Reductions	0	0	0	0	0	0	0	0	

L2 Cache

	Requests	Sectors	Sectors/Req	% Peak	Hit Rate	Bytes	Throughput	Sector Misses to Device	Sector Miss
L1/TEX Store	131072	524288	4	0.02	100	16777216	374824218.93	0	
L1/TEX Atomic ALU	0	0	0	0	0	0	0	0	
L1/TEX Atomic CAS	0	0	0	0	0	0	0	0	
L1/TEX Reduction	0	0	0	0	0	0	0	0	
L1/TEX Total	270440350	1081759668	4.00	36.49	98.66	34616309376	773372121104.67	14461268	
GCC Total	3010	12040	4	0.00	85.81	385280	8607642.36	1708	
ECC Total	-	0	-	0	-	0	0	0	
GPU Total	270460598	1081762265	4.00	36.49	98.66	34616392480	773373977753.11	14501988	

L2 Cache Eviction Policies

	First	Hit Rate	Last	Hit Rate	Normal	Hit Rate	Normal Demote	Hit Rate
L1/TEX Load	0	0	1080712680	98.71	524288	0	0	0
L1/TEX Store	0	0	0	0	524288	100	0	0
L1/TEX Atomic	0	0	0	0	0	0	-	-
L1/TEX Total	0	0	1080713640	98.71	1048576	50	0	0
GPU Total	74914	66.74	1080716404	98.71	1077914	51.20	0	0

Device Memory

	Sectors	% Peak	Bytes	Throughput
Load	14388560	1.13	460433920	10286675955.87
Store	574964	0.05	18398848	411053528.24
Total	14963524	1.17	478832768	10697729484.11

Scheduler Statistics

Active Warps Per Scheduler [warp]	2.82	No Eligible [%]	70.40
Eligible Warps Per Scheduler [warp]	0.45	One or More Eligible [%]	29.60
Issued Warp Per Scheduler	0.30		

Warps Per Scheduler

GPU Maximum Warps Per Scheduler
Theoretical Warps Per Scheduler
Active Warps Per Scheduler
Eligible Warps Per Scheduler
Issued Warp Per Scheduler
0.0
4.0
8.0
12.0
16.0

Warp State Statistics

Warp Cycles Per Issued Instruction [cycle]	9.52	Avg. Active Threads Per Warp	32
Warp Cycles Per Executed Instruction [cycle]	9.52	Avg. Not Predicated Off Threads Per Warp	31.05

Warp State (All Cycles)

Stall Math Pipe Throttle
Stall Wait
Selected
Stall Barrier
Stall Not Selected
Stall Short Scoreboard
Stall Dispatch Stall
Stall Long Scoreboard
Stall Branch Resolving
Stall No Instruction
Stall MIO Throttle
Stall LG Throttle
Stall IMC Miss
Stall Drain
Stall Misc
Stall Membar
Stall Sleeping
Stall Tex Throttle
Warp States
0.0
5.0
10.0
Cycles per Instruction

Instruction Statistics

Opcode Charts

Executed Instructions [inst]	5814046720	Avg. Executed Instructions Per Scheduler [inst]	17725752.20
Issued Instructions [inst]	5814055626	Avg. Issued Instructions Per Scheduler [inst]	17725779.35

Executed Instruction Mix

FMUL
IMAD
FFMA
HMMA
LOP3
FADD
FMNMX
LDSM
LEA
SHFL
IADD3
BRA
ISETP
MOV
F2FP
LDGSTS
LDS
UIADD3
BAR
SHF
BSYNC
BSSY
LDGDEPBAR
DEPBAR
S2R
CS2R
SEL
STG
LDG
UMOV
MUFU
ULDC
EXIT
Opcodes
0.0
500,000,000.0
1,000,000,000.0
1,500,000,000.0
Executed Warp-Level Instructions/Opcode

NVLink Topology

NVLink Topology

The system does not have any NVLink connections.

NVLink Tables

Logical NVLink Properties

The system does not have any NVLink connections.

NUMA Affinity

NUMA ID Table

GPU ID	GPU Name	NUMA ID by CPU Affinity	CPU Affinity	NUMA ID by Memory Affinity
0	NVIDIA GeForce RTX 3090	0	0-29	0

Launch Statistics

Grid Size	1024	Function Cache Configuration	CachePreferNone
Registers Per Thread [register/thread]	168	Static Shared Memory Per Block [Kbyte/block]	32.77
Block Size	128	Dynamic Shared Memory Per Block [byte/block]	0
Threads [thread]	131072	Driver Shared Memory Per Block [Kbyte/block]	1.02
Waves Per SM	4.16	Shared Memory Configuration Size [Kbyte]	102.40
Uses Green Context	0	Stack Size	1024
# SMs [SM]	82	# TPCs	41
Enabled TPC IDs	all	-	-

Occupancy

Theoretical Occupancy [%]	25	Block Limit Registers [block]	3
Theoretical Active Warps per SM [warp]	12	Block Limit Shared Mem [block]	3
Achieved Occupancy [%]	23.49	Block Limit Warps [block]	12
Achieved Active Warps Per SM [warp]	11.28	Block Limit SM [block]	16

Impact of Varying Register Count Per Thread

% Occupancy
125
100
75
50
25
0
0 8 16 24 32 40 48 56 64 72 80 88 96 104 112 120 128 136 144 152 160 168 176 184 192 200 208 216 224 232 240 248 256
Registers Per Thread

Impact of Varying Block Size

% Occupancy
125
100
75
50
25
0
0 32 64 96 128 160 192 224 256 288 320 352 384 416 448 480 512 544 576 608 640 672 704 736 768 800 832 864 896 928 960 992 1,024
Block Size

Impact of Varying Shared Memory Usage Per Block

% Occupancy
125
100
75
50
25
0
0 3,184 6,368 9,552 12,736 15,920 19,104 22,288 25,472 28,656 31,840 35,024 38,208 41,392 44,576 47,760 50,944 54,128 57,312 60,496 63,680 66,864 70,048 73,232 76,416 79,600 82,784 85,968 89,152 92,336 95,520 98,704 101,888
Shared Memory Per Block

Impact of Varying Register Count Per Thread

# Warps
44
33
22
11
0
0 8 16 24 32 40 48 56 64 72 80 88 96 104 112 120 128 136 144 152 160 168 176 184 192 200 208 216 224 232 240 248 256
Registers Per Thread

Impact of Varying Block Size

# Warps
35
26
18
9
0
0 32 64 96 128 160 192 224 256 288 320 352 384 416 448 480 512 544 576 608 640 672 704 736 768 800 832 864 896 928 960 992 1,024
Block Size

Impact of Varying Shared Memory Usage Per Block

# Warps
44
33
22
11
0
0 3,184 6,368 9,552 12,736 15,920 19,104 22,288 25,472 28,656 31,840 35,024 38,208 41,392 44,576 47,760 50,944 54,128 57,312 60,496 63,680 66,864 70,048 73,232 76,416 79,600 82,784 85,968 89,152 92,336 95,520 98,704 101,888
Shared Memory Per Block

GPU and Memory Workload Distribution

Average SM Active Cycles [cycle]	59894499.29	Average L1 Active Cycles [cycle]	59894499.29
Average L2 Active Cycles [cycle]	45541417.15	Average SMSP Active Cycles [cycle]	59891421.39
Average DRAM Active Cycles [cycle]	4987841.33	Total SM Elapsed Cycles [cycle]	5113166658
Total L1 Elapsed Cycles [cycle]	5113166658	Total L2 Elapsed Cycles [cycle]	2964915696
Total SMSP Elapsed Cycles [cycle]	20452666632	Total DRAM Elapsed Cycles [cycle]	5098634240

Workload Distribution

	Average	Min	Max	Sum
SM Active Cycles	59894499.29	57565273	62362218	4911348942
SMSP Active Cycle	59891421.39	57550665	62347402	19644386216
L1 Active Cycles	59894499.29	57565273	62362218	4911348942
L2 Active Cycles	45541417.15	45356702	45790296	2185988023
DRAM Active Cycle	4987841.33	4984928	4992128	59854096

Source Counters

Branch Instructions [inst]	123072512	Branch Efficiency [%]	100
Branch Instructions Ratio [%]	0.02	Avg. Divergent Branches [branches]	0

Warp Stall Sampling (All Samples)

Location	Value	Value (%)
0x75552d2873b0 in flash_at...	44,846	2
0x75552d284880 in flash_at...	41,265	2
0x75552d288990 in flash_at...	40,888	2
0x75552d2848c0 in flash_at...	40,633	2
0x75552d284eb0 in flash_at...	39,934	2

Most Instructions Executed

Location	Value	Value (%)
0x75552d286270 in flash_at...	2,797,568	0
0x75552d286260 in flash_at...	2,797,568	0
0x75552d286250 in flash_at...	2,797,568	0
0x75552d285ea0 in flash_at...	2,797,568	0
0x75552d285e90 in flash_at...	2,797,568	0

To customize your report even further, you might want to learn about custom sections and writing your own rules.
You might also want to consider adding individual metrics.